

• CLOSEST PAIR OF POINTS:

1D

Q. Given a set of points on a real line, find a pair of closest points.

Say input = $\{p_1, \dots, p_m\}$ where each $p_i \in \mathbb{R}$.

↳ provided in an arbitrary order

→ Sort them and record distances between consecutive points.

Pick the pair of consecutive points that attain minimum.

2D

 $\mathbb{R}^2$

Q. Given a set of points in a plane, find a pair that attains closest distance.

$\{p_1, \dots, p_n\}$ s.t. $p_i \in \mathbb{R}^2$

p_i, p_j
 (x_1, y_1) (x_2, y_2)

Trivial: Compute distances for all $\binom{n}{2}$ pairs and pick the min.

$\Rightarrow O(n^2)$

$$\left| \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \right|$$

* Sort by x and get cons. dist.

$P = \{p'_1, \dots, p'_n\}$

* Obtain the median x value

and divide the space into two parts: P_L and P_R .

* Obtain "recursively" closest pair of points in each of P_L and P_R .

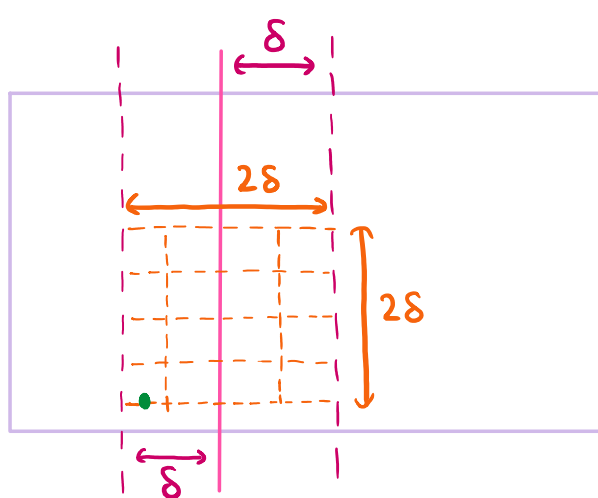
↓
keep dividing

$\delta_L =$ dist. attained by closest pair of points in P_L .

$\delta_R =$ " " " " P_R .

$$\delta = \min \{ \delta_L, \delta_R \}$$

Want to check whether any pair of points with one on each side of the partition attain a value smaller than δ .



Obs: This band could contain all the points $\{p_1, \dots, p_n\}$.

• Brute force for points in band:

↳ For all pairs of points, find their dist. and check if their min $\leq \delta$.

* Sort the points in the band w.r.t. their y -coordinates.

CLAIM: # of points in the $2\delta \times 2\delta$ grid ≤ 16

PROOF: It is sufficient to prove that each $\frac{\delta}{2} \times \frac{\delta}{2}$ square contains at most one point.

Suppose not and there are 2 points in a square of size $\frac{\delta}{2} \times \frac{\delta}{2}$.

$\Rightarrow$ Distance between them is at most $\sqrt{2} \times \frac{\delta}{2}$ (length of diagonal).

This contradicts that δ is min. of min. dist. across L and R .

From the above argument, we get that for each point in the band (point considered in the order of increasing y values), we only need to compare with next 15 points.

Record the min. over all the computations thus made.

Check if it is $< \delta$.

If yes: report the corr. points as closest set of points.

Else: report the points that achieve δ in L or R .

$$T(n) = 2 \cdot T\left(\frac{n}{2}\right) + 15n \Rightarrow O(n \log n) \text{ arithmetic operations}$$

+ Preprocessing of $O(n \log n)$
to maintain sorted list of x -values / y -values